Datathon 2026 | Monetizing Trust

Case Addendum: Revenue Assumptions, Data Dictionary, Judging, and Deliverables

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This addendum accompanies the case document. It gives you the assumptions you need to turn the datasets into dollars, defines every field, and sets out how your work will be scored and what you must submit. Read it before you start modeling.

1. Revenue Assumptions

Every team must use the assumptions below so that results are comparable across teams. If you depart from any of them, state the change and your reasoning explicitly in your submission.

1.1 Scaling to the full platform

The datasets contain 20,000 users drawn at random from a base of 3.2 million monthly active users. Each sampled user therefore represents 160 real users. Multiply any sample-level figure by 160 to estimate the platform-wide result.

1.2 Turning each strategy into revenue

Strategy	How to calculate monthly revenue
Strategy 1: Targeted Advertising	predicted_ad_revenue_per_month is what a user generates per month if you show them ads. Sum it across every user you choose to advertise to. You do not have to advertise to everyone.
Strategy 2: Aggregated Data Licensing	A user with a data_value_score of 100 licenses for $0.90 per month. Value scales linearly, so revenue per user = (data_value_score / 100) x $0.90.
Strategy 3: Premium Subscription	You set the monthly price. A user will only subscribe if their max_subscription_price is at or above the price you choose. Among those users, subscription_adoption_probability is the chance they actually subscribe. Revenue = sum of (adoption probability x your price) across eligible users. Subscribers get an ad-free experience, so a subscriber no longer generates advertising revenue. The overlap is small; you may ignore it if you say so.
Strategy 4: Brand Insight Partnerships	A user with a brand_insight_value_score of 100 is worth $0.55 per month once contracts are fully signed. Revenue per user = (brand_insight_value_score / 100) x $0.55. See the ramp schedule below.

Every formula in this table gives revenue before churn. Section 1.4 explains how churn reduces it over time.

1.3 The brand insight ramp

Advertising, data licensing, and subscription revenue may be assumed to start in month 1 at full level. Only brand insight partnerships follow the ramp below.

Brand insight partnerships are enterprise contracts and do not generate revenue immediately. Pulse would need to build the analytics product and sign brands one at a time. Assume the following share of full-ramp revenue is realized:

By month 3	By month 6	By month 8	By month 12
4%	11%	18%	100%

Between checkpoints, the share rises in a straight line from 0% at launch (month 0). From month 12 onward it stays at 100%.

Month	1	2	3	4	5	6	7	8	9	10	11	12+
Share of full revenue	1.3%	2.7%	4.0%	6.3%	8.7%	11.0%	14.5%	18.0%	38.5%	59.0%	79.5%	100%

This matters. Pulse has eight months of cash. Brand insight revenue counts in any month in which Pulse still has cash. On its own, though, it arrives too late to keep Pulse funded through month 8.

1.4 How churn works

Each churn field is the probability that a user leaves Pulse within the first three months after a strategy launches. A user who leaves is gone permanently and generates no revenue of any kind (advertising, subscription, licensing, or brand insight) after that.

Which churn applies to which user.

- ad_targeting_churn_risk applies to the users you choose to show ads to.
- data_sale_churn_risk applies to every user if Pulse licenses data, because users react to the policy rather than to their own record being sold.
- baseline_churn_probability applies to every other user. This includes all users under Premium Subscription and Brand Insight Partnerships. Those two strategies have no churn field of their own, so they use baseline churn. This puts every strategy on the same footing.
- A strategy churn field replaces baseline churn for that user. Never add the two together.
- If you show a user ads and Pulse also licenses data, add the two churn risks together for that user, up to a maximum of 1 (100%). This is the only case where two churn fields are added.

Timing. Churn happens once, spread evenly over the first three months. It does not repeat after month 3.

- In months 1, 2 and 3, a user's expected monthly revenue = full revenue x (1 - churn x m / 3), where m is the month number.
- From month 3 onward, a user's expected monthly revenue = full revenue x (1 - churn).

Example. A user generates $0.60 per month from ads and has an ad_targeting_churn_risk of 0.30. Month 1: $0.60 x (1 - 0.30 x 1/3) = $0.54. Month 2: $0.60 x (1 - 0.30 x 2/3) = $0.48. Month 3 onward: $0.60 x (1 - 0.30) = $0.42.

Counting users lost. Use these to describe how many users a strategy costs Pulse.

- Users lost under a strategy = sum of that strategy's churn field across the affected users, x 160.
- Extra users lost compared with doing nothing = sum of (strategy churn - baseline churn) across the affected users, x 160.
- Baseline churn is the do-nothing comparison point.

Some users have a strategy churn risk below their baseline. Use the values as given.

1.5 The financial target

Pulse holds $14 million in cash and burns $1.75 million per month. To reach 18 months of runway, monthly net burn must fall to about $777,778, which means new revenue of about $972,222 per month. This is where the case's $1 million per month figure comes from. Your recommendation should reconcile back to a runway figure, not just a revenue figure. The case says this revenue must start immediately, before current cash runs out. Read that against the eight months of cash: the revenue has to arrive before the money runs out.

The $972,222 figure assumes the revenue is at full level from month 1. Because churn (section 1.4) and the brand insight ramp (section 1.3) make revenue change from month to month, check your runway with the month-by-month method rather than the single monthly figure alone.

Month-by-month cash method.

- Start with $14,000,000 of cash.
- Each month: cash = last month's cash - $1,750,000 + that month's revenue.
- Runway = the number of months before cash falls below zero. A month that ends with exactly $0 still counts as funded.

Assume the monthly burn of $1.75 million stays fixed, and that no strategy adds operating costs, unless you choose to model costs and state your assumption.

1.6 Other assumptions

- **No user growth.** Model the platform as 3.2 million monthly active users minus churn. Do not add new users.
- **No interaction effects.** Strategies do not change each other's adoption or churn, except where this addendum says so. For example, offering a subscription does not reduce anyone's ad churn, subscribers still count toward data licensing and brand insight value, and subscribers still face data_sale_churn_risk if Pulse licenses data.
- **Overlap rule.** The subscription and advertising overlap rule in section 1.2 still applies: subscribers stop generating ad revenue, and you may ignore this if you say so.
- **No extra costs.** No strategy adds operating costs, unless you choose to model them and state your assumption.
- **Your own assumptions.** You may use a different assumption if you state it clearly and explain why, as the start of section 1 already allows.

2. Data Dictionary

All four files contain the same 20,000 users and join on user_id. The workbook all_users_combined.xlsx has them already joined, one row per user with every field, and opens on a Start here tab. Use it if you would rather not join the files yourself. Two scales are used throughout. Score fields run from 0 to 100, where higher always means more of the thing being measured. Probability fields run from 0 to 1. Dollar fields are monthly figures in US dollars.

2.1 user_profiles.csv

Field	Type and scale	What it means
user_id	Text	Unique identifier. Joins to all other files.
age_group	Category	One of 18-24, 25-34, 35-44, 45+.
region	Category	One of North America, Europe, Asia-Pacific, Latin America, Other.
account_age_months	Integer, months	Months since the user joined. Pulse launched three years ago, so valid values run from 1 to 36.
daily_minutes_spent	Decimal, minutes	Average minutes on the platform per day.
sessions_per_week	Integer	Separate visits to the platform in a typical week.
content_category_preference	Category	Primary interest area: Fitness, Gaming, Fashion, Lifestyle, or Creative.
engagement_score	Score, 0 to 100	Composite of how actively the user consumes and interacts with content.
influence_score	Score, 0 to 100	How much the user shapes the behaviour of others: reach, followers, and whether their activity pulls other users in.
privacy_sensitivity_score	Score, 0 to 100	How much the user cares about how their data is collected and used. Higher means more sensitive.
trust_score	Score, 0 to 100	How much the user currently trusts Pulse. This is trust the platform already has and can lose.
data_sharing_attitude	Category	Self-reported stance on data sharing: Restrictive, Cautious, Neutral, or Open. Survey-based, so it is directionally useful but noisier than privacy_sensitivity_score.

2.2 engagement_behavior.csv

Field	Type and scale	What it means
user_id	Text	Joins to the other files.
posts_per_week	Integer	Pieces of original content published per week.
likes_per_week	Integer	Likes given per week.
shares_per_week	Integer	Times the user shared content per week.
comments_per_week	Integer	Comments written per week.
content_views_per_day	Integer	Pieces of content viewed per day. This drives advertising inventory.
creator_interaction_score	Score, 0 to 100	How much the user engages directly with creators rather than passively consuming.
community_engagement_score	Score, 0 to 100	How embedded the user is in the interest communities on the platform.

2.3 monetization_potential.csv

Field	Type and scale	What it means
user_id	Text	Joins to the other files.
predicted_ad_revenue_per_month	Dollars per month	Advertising revenue this user would generate per month if shown ads. Modeled from their content consumption and advertiser demand for their interests.
subscription_willingness_score	Score, 0 to 100	How interested the user is in a paid tier, independent of price.
max_subscription_price	Dollars per month	The highest monthly price this user would accept. A value of 0.00 means the user would not pay for a subscription at any price. About a quarter of the base sits at or above $4.99.
data_value_score	Score, 0 to 100	How commercially valuable this user's behavioural data is to outside buyers.
brand_insight_value_score	Score, 0 to 100	How much this user contributes to aggregated trend insights sold to brands.

2.4 risk_and_churn_metrics.csv

Field	Type and scale	What it means
user_id	Text	Joins to the other files.
baseline_churn_probability	Probability, 0 to 1	Chance the user leaves within three months with no change to monetization.
ad_targeting_churn_risk	Probability, 0 to 1	Chance the user leaves within three months if shown targeted advertising.
data_sale_churn_risk	Probability, 0 to 1	Chance the user leaves within three months if Pulse licenses behavioural data.
subscription_adoption_probability	Probability, 0 to 1	Chance the user subscribes, given the price is one they would accept. Use together with max_subscription_price.
trust_loss_if_data_sold	Score, 0 to 100	How much trust this user would lose if Pulse licensed data. Distinct from churn: a user can stay and still lose faith, which shows up later in engagement.
trust_loss_if_ads	Score, 0 to 100	How much trust this user would lose if shown targeted advertising.

Note on messy data: these files reflect a real production export. Expect a small number of missing values and outliers. Handling them sensibly, and saying what you did, is part of the work.

3. How You Will Be Judged

Scoring is weighted 75 percent on analytical work and 25 percent on ethical reasoning. You do not need a perfect financial model to do well. Judges reward sound, honest reasoning. Putting numbers on how many users a strategy would drive away is the strongest evidence you can give, but if you cannot quantify it, a clear and well-argued case made in words still earns real credit. The best ethics sections grow out of the analysis rather than sitting on top of it as a paragraph of good intentions.

Criterion	Weight	What judges are looking for
Segmentation and user value	20%	Did you find the groups that actually matter, and can you show they are real rather than asserted? Are you clear about which users drive value and which carry risk?
Revenue modeling	25%	Is the arithmetic correct? Are your key numbers explained clearly enough to follow? Did you scale to the platform correctly?
Risk and user loss	20%	Did you think through what each strategy could cost Pulse in lost users and trust, not just what it earns? Numbers are strongest, but a clear argument in words earns credit too. Did you connect your recommendation back to the 18-month runway goal?
Communication and evidence	10%	Are the visualizations doing real work? Is the recommendation clear enough for a leadership team to act on?
Ethical reasoning	25%	Did you identify the genuine trade-offs around consent, transparency, and trust? How did you treat the users who carry the most risk? Would you be comfortable if Pulse published your recommendation in full?

You may recommend one strategy or a combination of up to two. A combination is not automatically better than a single strategy. Defend whichever you choose.

4. Deliverables and Timeline

Teams are 2 to 4 people. Every team submits the following.

- A slide deck, submitted as a PDF, maximum 10 slides excluding the title slide and any appendix. The appendix is not scored but judges may look at it.
- One named contact per team for finals scheduling.

How to submit.

- Email your slide deck as a single PDF to broadanalyticsassociation@gmail.com before Tuesday, October 20 at 11:59 PM.
- In the email, include your team name, the full names of all team members, and your named contact person with their email address.
- Name your file TeamName_Datathon2026.pdf.

When	What happens
Friday, October 16	Case, datasets, and this addendum released to all registered teams. The question window opens now and stays open until submissions close.
Tuesday, October 20, 11:59 PM	Submissions close. No late entries.
Wednesday, October 21	Internal judging. Finalists notified by the end of the day.
Wednesday, October 21, 6:00 to 8:00 PM, Room N21, Broad College of Business	Guest speaker session. Open to all participants, finalist or not.
Thursday, October 22, Entrepreneurship Lab, Broad College of Business	Finals. The exact time will be shared with finalist teams. Finalists present for 10 minutes followed by 5 minutes of questions from the judging panel. Awards follow.

Questions about the data or the case go to broadanalyticsassociation@gmail.com. Questions are accepted from release until submissions close on Tuesday at 11:59 PM. Any answer that affects the analysis will be sent to every team, not just the team that asked, so no one gains an advantage by asking. If something in the data looks wrong to you, ask rather than guessing.